Navigation and route planning

"PROVIDING USER AN ALTERNATIVE ROUTE DUE TO WATERLOGGING" *Generated by LeapSpace*

Waterlogging Alternative Routing at a Glance

Key Findings Table

Topic	Key finding	Affected users / context	Evidence
Detection foundation	Reliable waterlogging response depends on combining CCTV/edge-AI, IoT water-level/rainfall sensing, forecasts, and crowdsourced reports rather than relying on a single feed [1] [2] [3] [4] [5] [6] [7] [8]	Urban roads, flood-prone corridors, traffic operations	[1] [2] [3] [4] [5] [6] [7] [8]
Triggering reroutes	Practical rerouting triggers are usually threshold-based or prediction-based: water levels, forecast flood extent, or simple advisory states such as safe / precaution / do not traverse [8] [9] [10] [11]	Road users, emergency managers	[8] [9] [10] [11]
Network translation	Flood extent models can be translated into inundated links, blocked crossings, and inaccessible nodes for GIS-based diversion planning [12] [13] [14] [15] [16]	Transport agencies, routing engines	[12] [13] [14] [15] [16]
Routing logic	Real-time flood-aware routing typically uses GIS with graph/search algorithms such as A*, Dijkstra variants, bidirectional search, or Bayesian approaches to update paths as conditions change [17] [18] [19] [20] [21] [22]	Drivers, evacuees, responders	[17] [18] [19] [20] [21] [22]

Topic	Key finding	Affected users / context	Evidence
Safety vs efficiency	Safer routes often require detours; flood-avoidance routing is framed as improving resilience and exposure reduction, but typically adds travel time or distance, with one study reporting about 10-20% penalties [17] [19] [23] [24] [25] [26] [27]	Drivers, evacuees, urban networks	[17] [19] [23] [24] [25] [26] [27]
Communication	Driver compliance depends strongly on message design, placement, and detour credibility; VMS studies show meaningful diversion when guidance is clear and well positioned [28] [29] [30] [31] [32] [33]	Drivers approaching closures or flooded areas	[28] [29] [30] [31] [32] [33]
User-group differences	Evidence exists separately for motorists, pedestrians, evacuees, and emergency services, but there is little integrated comparison across groups in the same flooded-network setting [26] [34] [35] [36]	Multi-user transport planning	[26] [34] [35] [36]
Evaluation gap	Detection, routing, and communication are usually studied separately, so end-to-end operational evidence on latency, false alarms, route quality, and actual compliance remains weak [37] [38] [39] [40] [41] [42] [43]	System implementers	[37] [38] [39] [40] [41] [42] [43]

Direct Answer

To provide users with an alternative route due to waterlogging, the best-supported approach is an operational pipeline with four steps: **detect flooded links**, **convert detections into road-risk or closure flags**, **recalculate routes on a flood-constrained network**, and **communicate the detour through clear, credible messages** [1] [3] [8] [17] [19] [28] [31].

In practice, the route should usually optimize **safety and continuity rather than shortest travel time**, because flood-avoidance routes often require detours but reduce exposure to inundated roads and help preserve network functionality [17] [19] [23] [24] [25] [27]. A workable implementation would use sensor or forecast thresholds to flag risky links, verify them where possible with camera or crowdsourced evidence, then issue **mode-specific guidance**: detours for drivers, impassable-area warnings for pedestrians, and access-preserving routes for emergency vehicles [9] [10] [16] [34] [35].

Communication quality is not secondary: evidence from roadside message systems shows that users are more likely to follow rerouting advice when the message is credible, well placed, and paired with a reasonable alternative route [28] [29] [31] [33].

Confidence level: High - consistent multi-source evidence supports the detect-flag-route-communicate pipeline, though full end-to-end deployment evaluations are limited [1] [3] [8] [17] [28] [42].

Study Scope

This report covers research on waterlogging or flood-related transport disruption, emphasizing **2015-2025 operational studies and reviews** across urban flood detection, GIS/network modeling, dynamic routing, evacuation planning, and route communication systems [8] [9] [16] [17] [40]. Methods include CCTV and edge-AI sensing, IoT and forecast-based monitoring, social-media extraction, inundation mapping, graph-search routing, and communication/compliance studies centered mostly on VMS and in-vehicle guidance [3] [14] [28] [32] [38].

Assumptions & Limitations

The evidence base is fragmented: many studies assess either detection accuracy, flood mapping, or routing logic, but few evaluate the whole chain from automatic waterlogging detection through closure triggering, route guidance, and observed user outcomes [37] [38] [39] [42] [43]. Many routing models also simplify roads as fully open or fully closed, which understates partial passability, depth-dependent speed loss, evolving water levels, and detour congestion [14] [15] [43] [44] [45].

Suggested Further Research

- Evaluate full operational pipelines linking detection, trigger rules, rerouting, communication, and compliance in live urban waterlogging events [37] [42] [43].
- Develop routing models with partial passability, vehicle-specific water tolerance, and dynamic congestion rather than binary closures [14] [44] [45].
- Test whether predictive rerouting outperforms reactive rerouting during fast-changing pluvial flooding [8] [11] [21].
- Compare communication channels directly, especially mobile navigation, SMS, public alerts, and VMS for waterlogging-specific detours [28] [32] [46].
- Build integrated multi-user guidance for drivers, pedestrians, and emergency services in the same flooded-network scenario [34] [35] [36].

Full Report

Foundational Context

Providing alternative routes during waterlogging is a **network-constrained risk management problem** rather than a conventional shortest-path problem [17] [19]. The system must first determine whether a road segment is flooded or likely to flood, then translate that information into an operational state for each link: open, risky, or closed [8] [9] [10]. Once links are constrained, routing algorithms compute a new path that avoids inundated or high-risk segments while still preserving accessibility and keeping delays manageable [17] [18] [20] [21].

Three concepts are central:

1. **Detection**: identifying waterlogged roads from sensors, cameras, forecasts, or public reports [1] [3] [4] [5].
2. **Translation into network constraints**: mapping flood extent to road links, crossings, and nodes that become inaccessible or degraded [13] [14] [15] [16].
3. **User-facing diversion**: recalculating and communicating an alternative path that users will trust and follow [28] [31] [33].

Taken together, the literature suggests that the quality of the final diversion depends at least as much on **data fusion and communication design** as on the routing algorithm itself [8] [31] [33].

Confidence level: High - these concepts recur consistently across sensing, routing, and communication studies [1] [8] [17] [28].

1. Waterlogging Detection and Diversion Triggers

1.1 Sensing waterlogged roads from cameras, IoT, weather, and crowds

Real-time or near-real-time flood sensing is feasible through **CCTV/visual IoT, edge-AI, and sensor networks**, with event outputs available within minutes in some systems [1] [2] [47]. Camera-based systems are used to estimate flood depth, identify flood periods, or generate actionable location-tagged reports, although the provided evidence does not show many systems that directly and automatically trigger road closure status inside a traffic management workflow [37] [38] [39] [48].

Crowdsourced and social-media sources extend spatial coverage beyond official monitoring. Studies using Weibo, Facebook, internet media, and combined text-image recognition extracted road-adjacent waterlogging points and achieved substantial overlap with official records, including reported coverages of at least 79%, 82%, and 93% in separate cases [3] [7] [49]. A Changsha study also matched 51 flood points to official data while identifying 30 additional at-risk sites, suggesting that public reporting can surface events missed by formal channels [50].

IoT and forecast-linked systems contribute timeliness and predictive capability. Water-level, rainfall, and environmental sensing have been used to issue alerts and support emergency response decisions [4] [5] [6] [51]. FloNav is especially relevant because it combines ultrasonic sensing, forecasts, and a graph neural network to predict flooded areas and drive mobile rerouting [8]. Other systems use threshold exceedance or location-relative logic to issue user alerts, including expert systems that notify users when monitored water levels cross a predetermined threshold and road-monitoring prototypes that classify conditions into safe, precautionary, or never traverse [9] [10].

The main practical implication is that **no single detection channel is sufficient**. Cameras offer visual confirmation, IoT offers structured continuous measurements, forecasts support anticipation, and crowdsourcing expands coverage [1] [3] [4] [8].

Confidence level: High - multiple source types are consistently reported, and data fusion is one of the strongest cross-study themes [1] [3] [4] [5] [6] [8].

1.2 Trigger rules and prediction models for rerouting

Rerouting requires converting raw flood evidence into an actionable decision rule [8] [9] [10]. The evidence shows three main trigger styles:

- **Threshold-based triggers**: when water level exceeds a preset threshold, alerts are issued to nearby users or authorities [9] [52].
- **Advisory state models**: systems classify a road into discrete operational states such as safe, precautionary, or never traverse [10].
- **Prediction-based triggers**: forecasted flood extent or learned flood-risk models are used to reroute before conditions worsen [8] [11].

Predictive rerouting appears to be an emerging direction. FloNav couples live sensing with predicted flood extent, while StormSense combines water-level sensing with forecasting for autonomous alerts [8] [11]. This pattern implies a shift from purely reactive closure handling toward **preemptive avoidance of soon-to-be flooded links**, although comparative operational evidence remains limited [8] [21].

A limitation is that the current evidence rarely reports trigger latency, false-positive rates, or how often a trigger actually improves the final route decision [38] [39] [43].

Confidence level: Medium - trigger mechanisms are visible, but their comparative operational performance is underreported [8] [9] [10] [11].

1.3 Mapping inundated links and inaccessible nodes

For routing, flood information must be converted into **network geography**: which road links are inundated, which crossings are unsafe, and which nodes become isolated [12] [13] [16]. Foundational work used raster-based inundation modeling from DEM and SAR inputs and showed better cell-level inundation classification than simpler alternatives [12] [53]. More transport-specific work extends this into road hazard mapping, link importance analysis, and systemwide accessibility assessment [13] [15] [54].

High-resolution urban models combine 2D inundation outputs with road depth-impact assessment, enabling intra-urban street-level risk evaluation [14]. Distributed hydro-meteorological forecasting has also been applied to more than 2000 river-road crossing points to support traffic interruption decisions and safe access or evacuation-route selection [16].

Operationally, this means routing engines should not only know that "flooding exists," but exactly **which links are excluded, penalized, or critical** [13] [14] [16].

Confidence level: High - the mapping from flood extent to transport-network disruption is well supported [12] [13] [14] [15] [16].

2. Alternative Routing Algorithms Under Flooded-Road Constraints

2.1 Graph-search and dynamic rerouting methods

Real-time flood routing systems typically rely on **dynamic GIS-linked graph search**, including A*, Dijkstra variants, bidirectional Dijkstra, and Bayesian approaches [17] [18] [19] [20] [21] [22]. These methods continuously update feasible paths as flooding, accessibility, and traffic conditions change [17] [19] [21].

The practical routing logic is straightforward: remove or penalize flooded links, recalculate a feasible path, and refresh the recommendation when new conditions arrive [17] [18] [20]. Bayesian and other probabilistic approaches are especially useful when flood status is uncertain or evolving [22].

However, the supplied evidence does **not** provide strong head-to-head benchmarking of A* versus Dijkstra versus bidirectional Dijkstra under changing water levels [40] [41]. So while these methods are used, the literature here does not justify a claim that one is generally superior for waterlogging operations [40] [41].

Confidence level: Medium - algorithm families are clearly identified, but comparative performance evidence is thin [17] [18] [19] [20] [21] [22].

2.2 Flood-aware evacuation and emergency routing

Flood-aware routing is strongly framed as a **resilience and emergency-access tool** [17] [23] [24] [25]. In evacuation scenarios, safer routes bypass inundated segments and can reduce exposure or trapping risk compared with shortest-path routing, though they may increase travel time [26] [55]. For emergency services, flooding can reduce service coverage, lengthen rescue routes, and isolate vulnerable sites under severe rainfall and congestion [34].

For motorists more broadly, alternate routes are used to reduce flood exposure while preserving mobility [36]. For pedestrians, inundated streets reduce walkability and may render some areas impassable [35]. This implies that route generation should be **mode-specific**, not just road-user generic [34] [35] [36].

Confidence level: High - flood-aware routing's safety and access role is strongly supported across evacuation, motorist, and emergency-response contexts [17] [23] [24] [25] [26] [34].

2.3 Route optimality, robustness, and computation time

The evidence supports a clear pattern: flood-aware routing prioritizes **safe reachability and continuity**, often at the cost of extra time or distance [17] [19] [27]. One study estimated a typical 10-20% travel time or distance penalty for alternate routes, while noting that real-time rerouting could be worse if backtracking is required [27].

Several studies report that real-time information can improve route accuracy and traffic operations relative to static approaches [56] [57] [58] [59]. But the supplied excerpts do not provide a robust set of computation-time or robustness comparisons under dynamically rising water levels [40] [41].

Taken together, the evidence suggests that route quality should be judged with at least four criteria: **flood exposure avoided, added travel cost, continuity of access, and adaptability to changing conditions** [27] [42] [55].

Confidence level: Medium - the safety-detour trade-off is clear, but runtime and robustness evidence is limited [27] [40] [41].

3. Communicating Alternative Routes to Users

3.1 In-app navigation, alerts, and warning systems

Alternative-route communication improves safety by alerting users to flood risk and directing them to safer or less congested paths [19] [21] [22] [60]. Some systems include mobile or automated alert components, such as FloNav's app-based guidance and IoT systems that send alerts when thresholds are crossed [8] [9] [11].

However, the retrieved evidence does **not** include strong direct studies of waterlogging-specific detour guidance delivered through mainstream in-app navigation, SMS, or public emergency alert systems with measured user compliance [28] [30] [32] [46]. The behavioral evidence is stronger for roadside signs than for app-based flood rerouting [28] [31].

3.2 Roadside variable-message signs and compliance

The VMS literature is the strongest direct evidence on compliance. One field study reported high compliance, with about one in five vehicles changing route as recommended and almost none reaching the closed road section [28]. Other studies found that suggested-route or qualitative-delay messages increased diversion, while poorly designed panels reduced compliance [29] [30] [61]. Message content, sign placement, and detour quality repeatedly influenced whether users followed guidance [31] [32] [33].

For waterlogging operations, this means a message should ideally do three things: indicate that the current road is unsafe, offer a credible alternative, and present the diversion early enough for action [28] [31] [33].

3.3 User adherence and operational response

Route-diversion studies using connected-vehicle data reported diversion rates of 58% to 93% during long closures and recommended real-time monitoring so operators can adjust diversion resources [62]. This is not waterlogging-specific, but it shows that agencies should monitor uptake and modify operations when diversions saturate or compliance drops [33] [62].

The broader lesson is that **good routing logic is not enough**; agencies also need credible messaging and feedback on whether users are actually diverting [28] [33] [62].

Confidence level: High - the role of communication design is consistently supported, though evidence is strongest for VMS rather than mobile flood-alert channels [28] [29] [31] [33].

4. Safety, Accessibility, and User-Group Differences

Waterlogging affects traveler groups differently. For **drivers**, rerouting reduces exposure to flooded links but introduces travel disruption and possible unfamiliarity with detours [36] [45]. For **pedestrians**, inundation degrades walkability and can make some areas impassable [35]. For **emergency services**, flooding can cut coverage, delay rescue access, and isolate vulnerable destinations [34].

The retrieved papers do not provide a single integrated comparison across drivers, pedestrians, cyclists, and emergency vehicles in the same scenario [26] [34] [35] [36]. Still, taken together, these findings suggest that effective rerouting should use **different passability thresholds and priorities by user type** [34] [35] [36].

Inferred: A unified operational system would likely perform better if it differentiated route rules for low-clearance vehicles, pedestrians, and emergency vehicles rather than using a single flooded-road rule for all users [34] [35] [36].

Confidence level: Medium - direct evidence for user differences exists, but integrated multi-group system evaluations are limited [34] [35] [36].

5. Evaluation Metrics and Reported Limitations

5.1 Performance metrics for diversion systems

The literature evaluates flood-aware diversion using several families of metrics:

- **Detour cost**: added travel time and distance [27].
- **Safety / incident avoidance**: fewer trapped evacuees or fewer traversed flooded segments [55].
- **Accessibility and resilience**: preservation of network access, reduced isolation, or improved recovery [42] [43].
- **Detection-oriented metrics**: classification accuracy and overlap with official reports for sensing systems [3] [7] [38].

This mix shows that current evaluations are often **partial**, focusing either on sensing accuracy or on modeled network effects rather than full operational user outcomes [38] [42].

5.2 False alerts, partial failures, and model limits

False-alarm handling is weakly covered. One study used a zero-point offset and a 90th-percentile threshold to determine flood periods from CCTV-derived depth signals, but broader false-positive/false-negative reporting is uncommon in the provided evidence [39]. Detection studies often report classification accuracy, such as 92.45% test accuracy in one classifier, without translating that into road-closure decision quality [38].

Routing studies often simplify network disruption as **binary link failure**, overlooking partial passability, evolving depth, vehicle-specific tolerance, and congestion spillback [14] [15] [43] [44] [45]. This limits realism for urban waterlogging, where roads may degrade progressively rather than close instantly [43] [44].

5.3 Detour overload and user noncompliance

Motorists may ignore alternative routes when they are unfamiliar or seem unconvincing [45]. Communication studies show that poor sign design or weakly justified detours reduce adherence, while strong route guidance increases diversion [28] [33] [61]. Route-diversion monitoring is therefore important not just for user information but for network control, because overloaded detours can undermine the benefits of rerouting [45] [62].

Confidence level: High - limitations around fragmentation, simplified disruption modeling, and communication-dependent adherence are strongly recurrent [33] [39] [43] [44].

Practical Implementation Pattern

A practical evidence-based design for providing alternative routes during waterlogging would be:

1. **Fuse detection inputs** from cameras, IoT sensors, rainfall/water forecasts, and crowdsourced reports [1] [3] [4] [8].
2. **Assign road states** using thresholds or prediction outputs: open, caution, or closed/high-risk [8] [9] [10].
3. **Map states to network links and nodes** using GIS and inundation/road-impact models [14] [15] [16].
4. **Recalculate routes dynamically** with flood-constrained graph search [17] [18] [19] [20] [21].
5. **Communicate the detour clearly** through VMS, mobile apps, or alerts, explaining the hazard and recommending a credible alternative [28] [31] [33].
6. **Monitor diversion uptake and network effects** so detour resources and messages can be adjusted if compliance is low or alternate routes become overloaded [45] [62].

Inferred: A confidence-scored approach could improve operations by distinguishing between warning-only, soft-avoidance, and hard-closure routing depending on how strongly multiple data sources agree on flooding risk [8] [9] [22].

Confidence level: Medium - each step is evidenced separately, but the full integrated pipeline is not yet well validated in one deployment [37] [42] [43].

Conclusion

The strongest evidence supports treating waterlogging rerouting as a **multi-source, near-real-time decision system**: detect inundation from cameras, sensors, forecasts, and crowdsourcing; convert that evidence into road constraints; compute safer routes on a dynamic network; and communicate those routes with credible, well-timed messages [1] [3] [8] [16] [17] [28]. The main operational trade-off is clear: safer routes usually cost extra time or distance, but they reduce exposure to flooded links and improve network resilience [17] [19] [26] [27].

The main weakness in the current evidence base is not the lack of useful components, but the lack of **end-to-end validation** connecting detection quality, trigger logic, route quality, message design, and actual user behavior in live waterlogging operations [38] [39] [41] [43].

Overall confidence: High - the core architecture and trade-offs are strongly supported across studies, though full-system operational proof remains limited [1] [3] [8] [17] [33] [43].

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